SAGE-Extend 3D: pipeline repair, geometry-aware knowledge-grounded editing, and a controlled pilot (interim technical report)

SAGE-Extend 3D project (generated from repository artifacts)

2026-10-14 15:51

Read first. Only the input photographs are real. Every 3D asset and every rendered view is *synthetic*. SAGE contains a single view per specimen, so **no novel view, hidden structure or 3D shape is verified against reality**. Biological plausibility was **not** assessed by experts (none available). All numbers come from a 15-specimen pilot on a local 2×TITAN X machine. Effects are reported with confidence intervals; nothing is claimed significant unless stated.

Summary

- **Reconstruction.** The low input-view IoU reported in the interim was a SAM 3D pose/scale error (objects over-scaled in 13/15), not a camera error. A pose-only correction raises median IoU from 0.63 to 0.92 on all 15 reconstructions. This is a fit to the input silhouette, not an accuracy measurement. One specimen (Peach 0006) is a reconstruction failure.
- **Arm F** failed only because of prompt serialisation (fixed and verified on live runs). A second bug (non-finite opacity → NaN → every fp16 step skipped) explained the "no change" on Peach 0006.
- **GaussianEditor does not edit geometry.** Its Gaussians move incoherently (displacement coherence 0.31 vs 0.25 for noise); the sheet thickens and the texture becomes grainy. KB text changes its colour output but not its geometry: F vs E coherence 0.31 vs 0.31 (Δ +0.00, specimen-bootstrap 95% CI [+0.00, +0.00], 11/15 specimens favour the first arm, 5 classes).
- **A KB-driven geometry editor** produces coherent, measurable deformation (coherence ≈1). Against an equal-magnitude generic warp, KB geometric recall is 0.44 vs 0.07 (Δ +0.37, specimen-bootstrap 95% CI [+0.13, +0.61], 4/6 specimens favour the first arm, 2 classes). Recall is partly by construction; biological correctness is unvalidated.
- **Cross-view consistency** (same surface point, adjacent views, median ΔE): per-view 2D edits 16.8/19.3 (generic/KB), GaussianEditor 3D 9.3, geometry-only 3D 2.0, against a floor of 2.0. This is internal consistency only.
- **Not established:** external validity of any novel view (no reference data), biological validity (no experts) and downstream utility (not run).

1 Data, scope and provenance

All inputs are single real photographs from SAGE (HF `tirtho149/SAGE`, CC-BY-4.0), from the leakage-safe *build* split (SHA-1 dedup, pHash clusters, grouped splits; `reports/phase1_protocol/02_dataset_protocol.md`). Fifteen pilot specimens span five classes: Peach Leaf Curl, Tomato Leaf Mold, Corn Smut, Citrus Canker and Mango Anthracnose (negative control). **No second real view or 3D measurement of any specimen exists.** By user decision, no external multi-view dataset was used. Every rendered view of a 3D asset is therefore *synthetic and unverified*. Expert review was not available. All runs used the local machine (2× GTX TITAN X, 12 GB, Maxwell); Nova/Slurm were not used.

2 Methods

Reconstruction and alignment. SAM 3D Objects (stage-wise CPU/GPU offload; locally reproduces the A100 output to a Chamfer distance of 4.5×10^{-5} of the bbox diagonal). Input-view alignment study A0–A3: A0 is the existing pose; A1 tests the camera convention; A2 is a closed-form scale and translation; A3 is a 7-DoF render-and-compare of the object pose with geometry frozen and a mask-only loss. The corrected pose is A3* = A3 if it beats A2, otherwise A2. LPIPS and MoGe depth agreement are held out.

GaussianEditor diagnosis. Code inspection plus per-Gaussian before/after comparison, using prune-aware index correspondence (verified bit-exact on a no-op run): displacement, k-NN displacement coherence (1 = smooth field, 0.25 = i.i.d. noise), rotation, scale, opacity, colour, sheet thickness.

Geometry editor (G2). Edited positions $\mu_i' = \mu_i + m_i d_i$, with d minimising $\lambda_1 \operatorname{mean}_i m_i (d_i \cdot n_i - h_i)^2 / L^2 + \lambda_2 \operatorname{mean}_{ij} \big((\|e_{ij}'\| - \|e_{ij}\|)/\|e_{ij}\| \big)^2 + \lambda_3 \operatorname{mean}_{ij} \|m_i d_i - m_j d_j\|^2 / L^2 + \lambda_4 \operatorname{mean}_i (1 - m_i) \|d_i\|^2 / L^2$, with $(\lambda_1..\lambda_4) = (1, 0.1, 0.01, 1)$ over an 8-NN graph. L is the bbox diagonal, so all quantities are unit-free. The target normal offset h is built from KB operators: *pucker* (zero-mean Poisson-disk blisters), *distort* (low-frequency warp), *thicken* (Gaussian scale along the normal), *outgrowth* (galls), *depress* (pits). Rotations follow the change of the local normal. Colours and opacities are untouched. The generic control G uses a random smooth vector field of equal RMS.

Structured KB grounding. Only the verbatim web quote is evidence. It is split into stages at temporal markers, and one stage is edited. Prompts are a fixed template plus verbatim phrases (asserted in code). Hedged phrases are excluded and other-stage features are prohibited. Terms without an operator (curl, roll, wilt, corky) are recorded, never approximated.

Arms. A real; B generic 2D; C KB 2D (interim mixed-stage prompt); C-NG KB without geometry; C-MM mismatched KB; C-stage stage-consistent KB 2D; I 2D shape-capable (mask dilated 15 %, image guidance 1.2); Bmv/Cmv independent per-view 2D edits of the ring renders (object-masked); D unedited 3D; E/F GaussianEditor generic/KB (interim settings); G generic geometry; J KB geometry; K GaussianEditor appearance-only with geometry frozen; H = J then K. GaussianEditor prompts are passed through validated per-run configs and checked against GaussianEditor's own parsed config after the run.

Metrics. Geometry: displacement and coherence; band-split out-of-plane attributes (distortion, blister density, normal thickening, outgrowth; the depression detector proved non-specific and is excluded from recall). Thresholds come from a uniform midpoint rule on a synthetic flat sheet; the operator×detector matrix is diagonal. KB recall is the fraction of the stage's required geometric attributes detected; the unsupported rate is the fraction of detected attributes not called for by the KB. Appearance: CIELAB term detectors with an a-priori threshold of 5 units. Render QC: silhouette and depth change, grain, edge fuzz. Cross-view: same-surface-point ΔE between adjacent views (depth-tested projections), with the unedited asset as the floor. Statistics: the specimen is the unit (seeds averaged); paired cluster bootstrap over disease classes, Wilcoxon, Holm.

3 Results

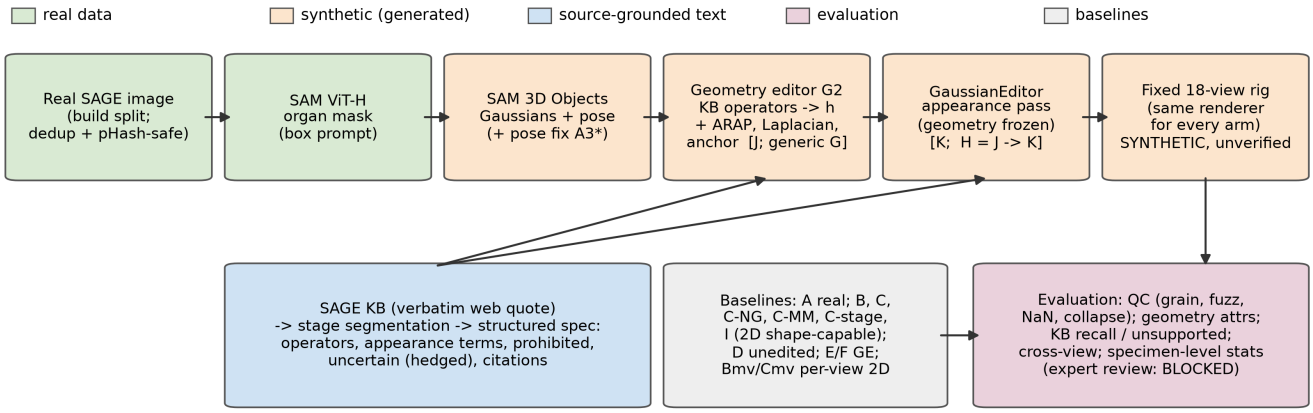

Figure 1: Pipeline and data status.

3.1 Input-view alignment (E1)

Table 1: E1 input-view alignment (IoU). A1 = principal point at centre; A2 = closed-form scale+translation; A3 = 7-DoF render-and-compare (mask loss only, geometry frozen); A3* = A3 if better than A2. † diverged. LPIPS and depth agreement with MoGe are never optimised.

Specimen	K px	A0	A1	A2	A3	A3*	area err	LPIPS (held-out)	depth err (held-out)	label
Citrus Canker 0004	0.6	0.67	0.67	0.84	0.91	**0.91**	0.44	0.29→0.30	0.025→0.079	ok
Citrus Canker 0012	0.3	0.73	0.73	0.98	0.99	**0.99**	0.36	0.23→0.21	0.095→0.029	ok
Citrus Canker 0016	5.6	0.76	0.77	0.98	0.98	**0.98**	-0.19	0.18→0.19	0.088→0.071	ok
Corn Smut 0012	0.4	0.63	0.63	0.86	0.94	**0.94**	0.43	0.20→0.18	0.011→0.015	ok
Corn Smut 0025	4.3	0.61	0.61	0.77	0.76	**0.77**	-0.09	0.22→0.21	0.039→0.042	ok
Corn Smut 0052	1.0	0.72	0.72	0.83	0.90	**0.90**	0.34	0.16→0.16	0.051→0.091	ok
Mango Anthracnose 0044	0.2	0.70	0.70	0.87	0.93	**0.93**	0.27	0.14→0.12	0.009→0.081	ok
Mango Anthracnose 0049	0.3	0.77	0.78	0.85	0.95	**0.95**	0.19	0.11→0.10	0.016→0.039	ok
Mango Anthracnose 0094	0.3	0.63	0.63	0.82	0.86	**0.86**	0.57	0.09→0.13	0.035→0.060	ok
Peach Leaf_Curl 0006	1.6	0.43	0.43	0.50	0.13†	**0.50**	0.68	0.18→0.20	0.031→0.027	reconstruction_failure
Peach Leaf_Curl 0063	6.4	0.58	0.58	0.67	0.00†	**0.67**	0.35	0.11→0.11	0.015→0.014	ok
Peach Leaf_Curl 0087	1.6	0.84	0.85	0.89	0.95	**0.95**	0.08	0.05→0.04	0.003→0.012	ok
Tomato Leaf_Mold 0015	0.3	0.40	0.40	0.88	0.94	**0.94**	1.50	0.20→0.21	0.017→0.031	ok
Tomato Leaf_Mold 0052	0.4	0.55	0.55	0.72	0.92	**0.92**	0.79	0.29→0.28	0.034→0.043	ok
Tomato Leaf_Mold 0092	0.5	0.47	0.47	0.84	0.23†	**0.84**	1.11	0.32→0.32	0.030→0.028	ok

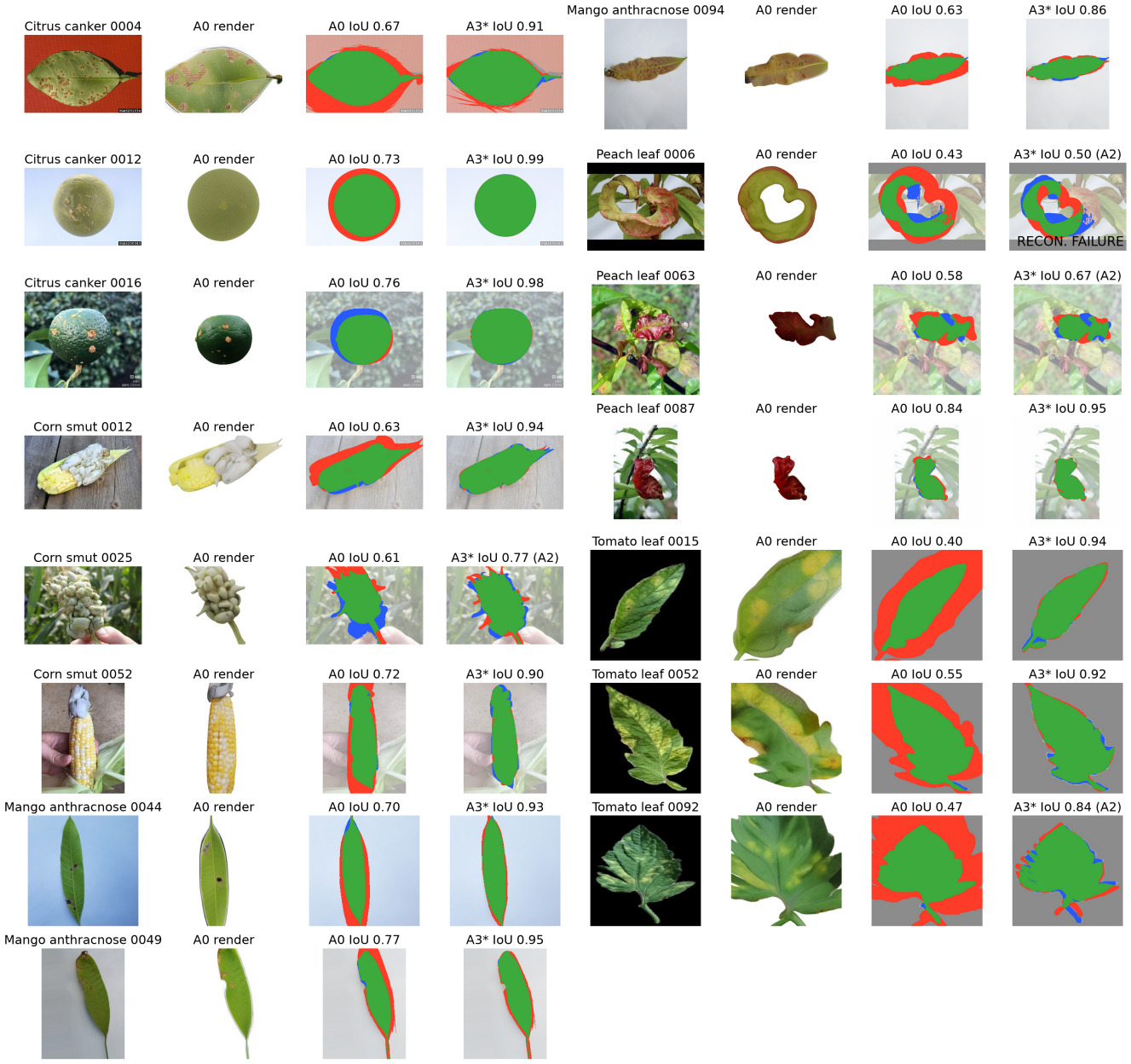

Figure 2: Reconstruction debugging: A0 vs corrected pose A3*. Fitted, not verified.

3.2 What GaussianEditor changes

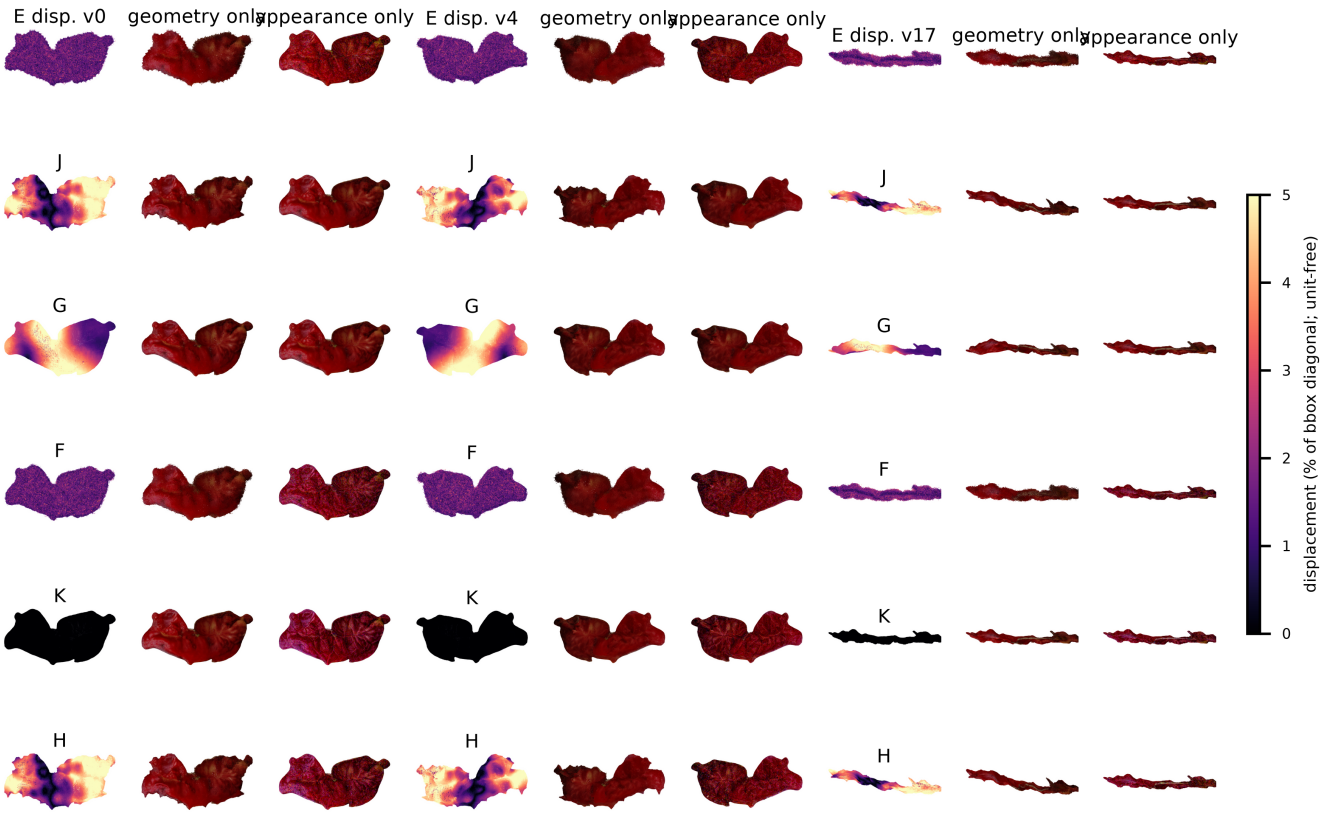

Figure 3: Displacement heatmaps and geometry-only / appearance-only decomposition (Peach 0087).

3.3 Geometry-aware editing and knowledge ablation

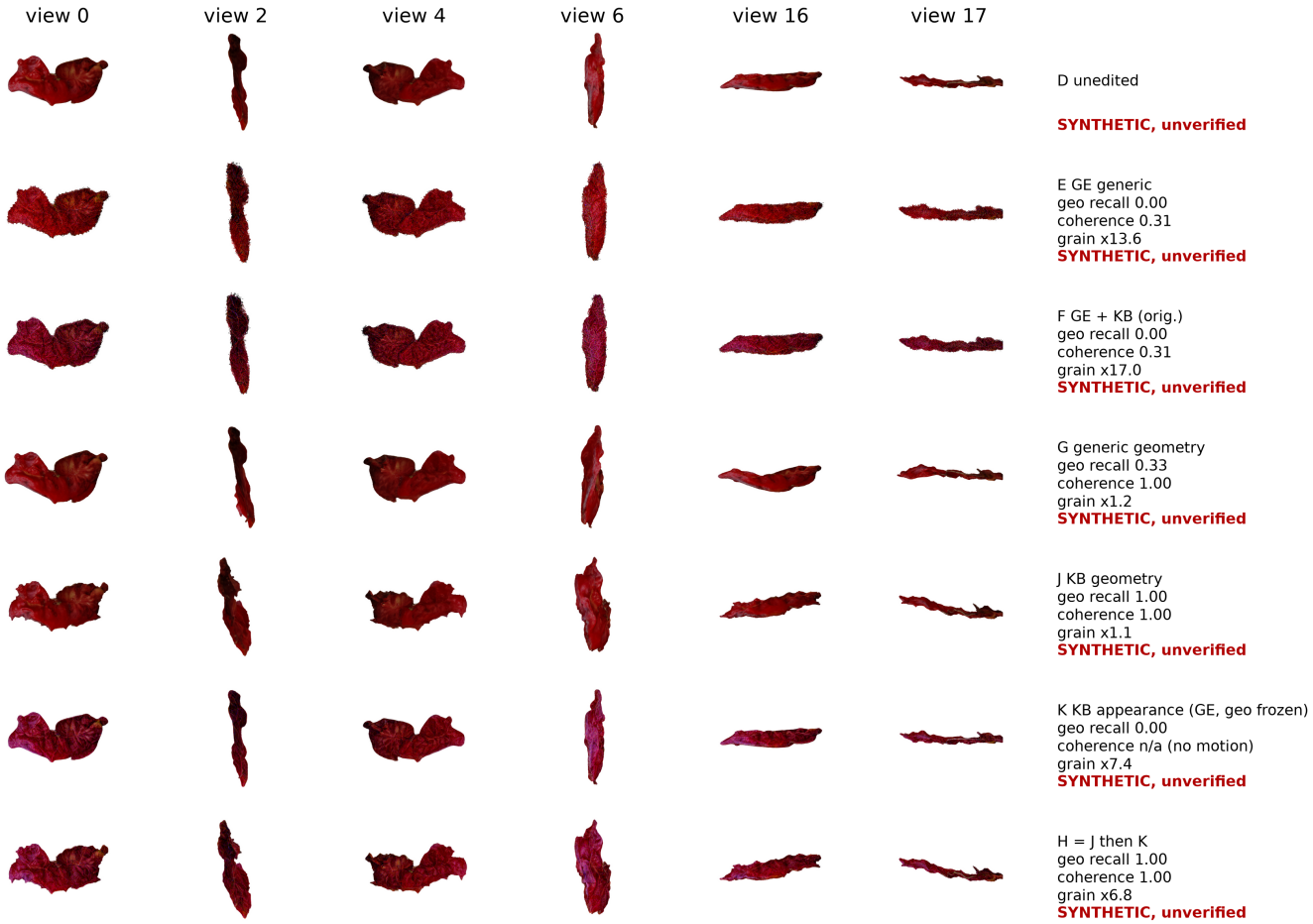

Figure 4: All 3D arms of Peach 0087 on fixed rig views (synthetic, unverified).

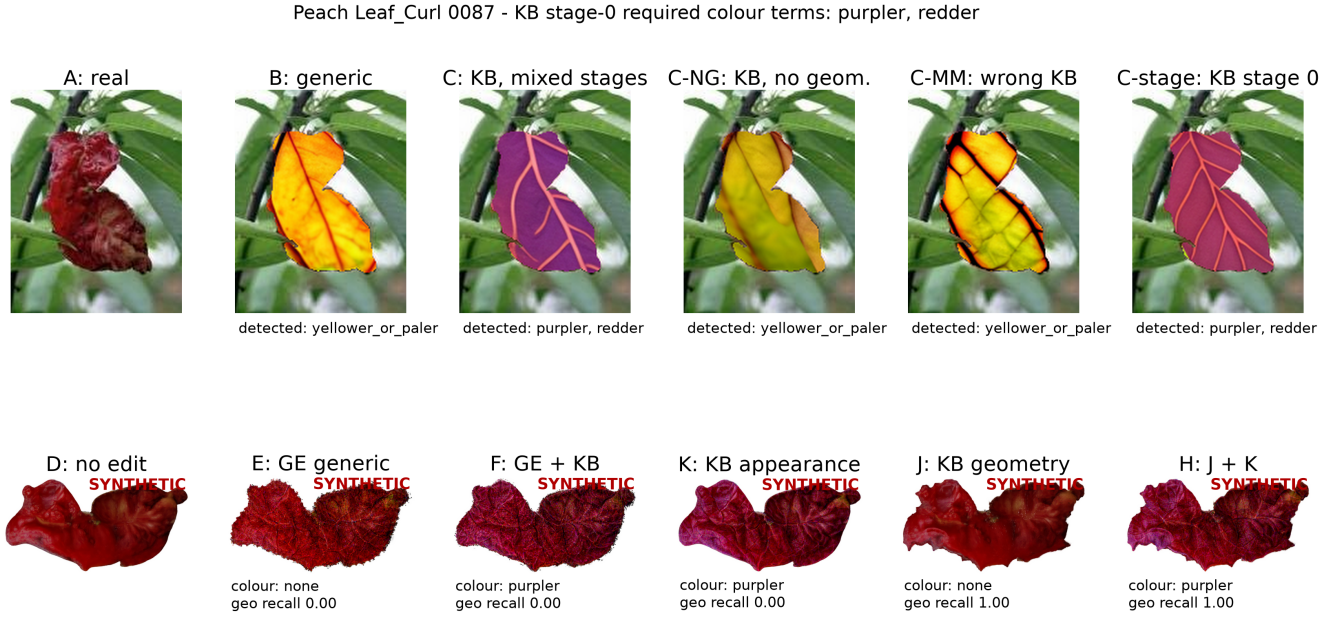

Figure 5: Knowledge ablation in 2D and 3D on Peach 0087.

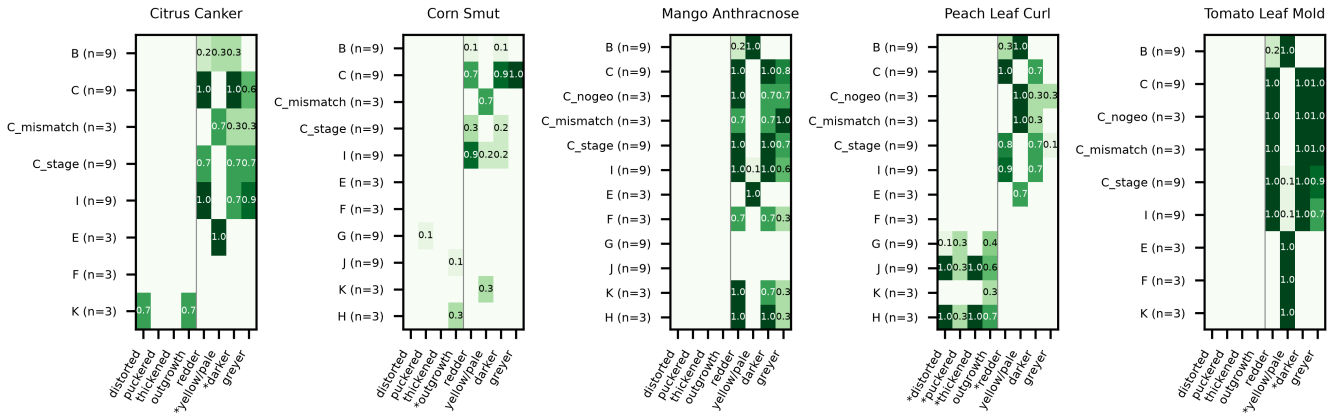

Figure 6: Requested (KB) vs realised attributes per class.

Table 2: Per-specimen metrics (seeds averaged). Recalls over specific detectors / scorable colour terms; – = not applicable or not scorable.

Specimen	Arm	seeds	geo recall	app recall	unsupp. geo	coherence	sil. change	grain	fuzz	ΔE
Citrus Canker 0004	D	1	–	0.00	0.00	–	0.00	1.0	1.0	0.0
Citrus Canker 0004	E	1	–	0.50	0.00	0.31	0.17	2.9	1.6	22.0
Citrus Canker 0004	F	1	–	0.00	0.00	0.31	0.18	3.3	1.5	16.5
Citrus Canker 0004	K	1	–	0.00	0.00	0.00	0.00	2.8	1.0	9.4
Citrus Canker 0012	D	1	–	0.00	0.00	–	0.00	1.0	1.0	0.0
Citrus Canker 0012	E	1	–	–	0.00	0.32	0.06	70.8	2.2	21.2
Citrus Canker 0012	F	1	–	–	0.00	0.32	0.06	78.4	2.3	18.4
Citrus Canker 0012	K	1	–	–	1.00	0.96	0.00	27.4	1.0	11.7
Citrus Canker 0016	D	1	–	0.00	0.00	–	0.00	1.0	1.0	0.0
Citrus Canker 0016	E	1	–	–	0.00	0.31	0.06	12.9	2.3	17.4
Citrus Canker 0016	F	1	–	–	0.00	0.32	0.07	15.2	2.3	15.7
Citrus Canker 0016	K	1	–	–	1.00	0.97	0.00	6.9	1.0	9.5
Corn Smut 0012	D	1	0.00	0.00	0.00	–	0.00	1.0	1.0	0.0
Corn Smut 0012	E	1	0.00	–	0.00	0.31	0.09	25.8	2.1	13.9
Corn Smut 0012	F	1	0.00	–	0.00	0.31	0.09	30.2	2.0	16.2
Corn Smut 0012	G	3	0.00	–	0.00	0.99	0.10	1.1	1.3	5.9
Corn Smut 0012	J	3	0.00	–	0.00	0.93	0.14	1.5	1.4	4.0
Corn Smut 0012	K	1	0.00	–	0.00	0.12	0.00	10.1	1.0	10.1
Corn Smut 0012	H	1	0.00	–	0.00	0.93	0.15	12.4	1.5	11.8
Corn Smut 0025	D	1	0.00	0.00	0.00	–	0.00	1.0	1.0	0.0
Corn Smut 0025	E	1	0.00	–	0.00	0.31	0.10	11.0	1.8	15.1
Corn Smut 0025	F	1	0.00	–	0.00	0.31	0.11	13.1	1.9	16.8
Corn Smut 0025	G	3	0.00	–	0.33	1.00	0.13	1.1	1.3	7.5
Corn Smut 0025	J	3	0.00	–	0.00	0.92	0.13	1.4	1.6	4.9
Corn Smut 0025	K	1	0.00	–	0.00	0.00	0.00	3.6	1.0	11.2
Corn Smut 0025	H	1	0.00	–	0.00	0.94	0.12	4.6	1.6	12.7
Corn Smut 0052	D	1	0.00	0.00	0.00	–	0.00	1.0	1.0	0.0
Corn Smut 0052	E	1	0.00	–	0.00	0.31	0.11	9.6	1.8	16.7
Corn Smut 0052	F	1	0.00	–	0.00	0.32	0.10	13.0	1.8	18.8
Corn Smut 0052	G	3	0.00	–	0.00	0.95	0.12	1.1	1.2	11.6
Corn Smut 0052	J	3	0.33	–	0.00	0.83	0.14	1.2	1.3	8.7
Corn Smut 0052	K	1	0.00	–	0.00	0.59	0.00	5.1	1.0	11.7
Corn Smut 0052	H	1	1.00	–	0.00	0.83	0.14	5.9	1.3	16.4
Mango Anthracnose 0044	D	1	–	0.00	0.00	–	0.00	1.0	1.0	0.0
Mango Anthracnose 0044	E	1	–	–	0.00	0.32	0.09	6.3	1.8	20.2
Mango Anthracnose 0044	F	1	–	–	0.00	0.32	0.09	7.6	1.6	14.9
Mango Anthracnose 0044	G	3	–	–	0.00	0.85	0.04	0.8	1.2	2.5
Mango Anthracnose 0044	J	3	–	–	0.00	0.60	0.04	0.8	1.3	2.6
Mango Anthracnose 0044	K	1	–	–	0.00	0.00	0.00	6.1	1.0	12.1
Mango Anthracnose 0044	H	1	–	–	0.00	0.60	0.04	5.2	1.3	13.2
Mango Anthracnose 0049	D	1	–	0.00	0.00	–	0.00	1.0	1.0	0.0
Mango Anthracnose 0049	E	1	–	–	0.00	0.31	0.20	10.4	2.0	17.4
Mango Anthracnose 0049	F	1	–	–	0.00	0.31	0.20	11.7	1.9	19.6
Mango Anthracnose 0049	G	3	–	–	0.00	0.95	0.04	1.0	1.1	2.8
Mango Anthracnose 0049	J	3	–	–	0.00	0.63	0.03	0.9	1.1	2.2
Mango Anthracnose 0049	K	1	–	–	0.00	0.00	0.00	7.8	1.0	16.1
Mango Anthracnose 0049	H	1	–	–	0.00	0.63	0.04	7.4	1.1	16.4
Mango Anthracnose 0094	D	1	–	0.00	0.00	–	0.00	1.0	1.0	0.0
Mango Anthracnose 0094	E	1	–	–	0.00	0.31	0.14	4.7	1.4	24.6
Mango Anthracnose 0094	F	1	–	–	0.00	0.32	0.14	5.9	1.3	17.9
Mango Anthracnose 0094	G	3	–	–	0.00	0.48	0.03	1.0	1.0	2.5
Mango Anthracnose 0094	J	3	–	–	0.00	0.45	0.02	1.0	1.1	1.7
Mango Anthracnose 0094	K	1	–	–	0.00	0.43	0.00	4.0	1.0	13.8
Mango Anthracnose 0094	H	1	–	–	0.00	0.45	0.02	3.8	1.1	14.1
Peach Leaf_Curl 0006	D	1	0.00	0.00	0.00	–	0.00	1.0	1.0	0.0
Peach Leaf_Curl 0006	E	1	0.00	0.00	0.00	0.32	0.12	20.5	2.2	22.1
Peach Leaf_Curl 0006	F	1	0.00	0.00	0.00	0.32	0.12	24.2	2.2	21.6
Peach Leaf_Curl 0006	G	3	0.00	0.00	1.00	0.98	0.21	1.2	1.3	10.2
Peach Leaf_Curl 0006	J	3	0.67	0.00	0.33	0.98	0.23	1.7	2.9	11.0
Peach Leaf_Curl 0006	K	1	0.00	0.00	1.00	0.97	0.00	11.0	1.0	17.2
Peach Leaf_Curl 0006	H	1	0.67	0.00	0.33	0.98	0.23	9.7	2.9	21.4
Peach Leaf_Curl 0063	D	1	0.00	0.00	0.00	–	0.00	1.0	1.0	0.0
Peach Leaf_Curl 0063	E	1	0.00	0.00	0.00	0.32	0.11	14.6	1.9	16.4
Peach Leaf_Curl 0063	F	1	0.00	1.00	0.00	0.32	0.11	19.0	1.9	19.9
Peach Leaf_Curl 0063	G	3	0.33	0.00	0.00	0.97	0.20	1.1	1.2	12.1
Peach Leaf_Curl 0063	J	3	0.78	0.00	0.22	0.96	0.22	1.9	2.4	12.7
Peach Leaf_Curl 0063	K	1	0.00	1.00	0.00	0.92	0.00	8.4	1.0	15.9
Peach Leaf_Curl 0063	H	1	0.67	1.00	0.33	0.96	0.23	8.1	2.5	21.0
Peach Leaf_Curl 0087	D	1	0.00	0.00	0.00	–	0.00	1.0	1.0	0.0
Peach Leaf_Curl 0087	E	1	0.00	0.00	0.00	0.31	0.14	13.6	1.9	13.5
Peach Leaf_Curl 0087	F	1	0.00	1.00	0.00	0.31	0.14	17.0	1.9	16.2
Peach Leaf_Curl 0087	G	3	0.11	0.00	0.17	1.00	0.23	1.1	1.1	9.0
Peach Leaf_Curl 0087	J	3	0.89	0.00	0.00	1.00	0.26	1.1	1.4	9.6
Peach Leaf_Curl 0087	K	1	0.00	1.00	0.00	0.00	0.00	7.4	1.0	12.3
Peach Leaf_Curl 0087	H	1	1.00	1.00	0.00	1.00	0.28	6.8	1.4	15.9
Tomato Leaf_Mold 0015	D	1	–	0.00	0.00	–	0.00	1.0	1.0	0.0
Tomato Leaf_Mold 0015	E	1	–	0.50	0.00	0.31	0.18	17.5	1.8	22.1
Tomato Leaf_Mold 0015	F	1	–	0.50	0.00	0.31	0.17	19.7	1.9	19.1
Tomato Leaf_Mold 0015	K	1	–	0.50	0.00	0.31	0.00	11.3	1.0	15.4
Tomato Leaf_Mold 0052	D	1	–	0.00	0.00	–	0.00	1.0	1.0	0.0
Tomato Leaf_Mold 0052	E	1	–	0.50	0.00	0.30	0.18	16.5	1.8	22.9
Tomato Leaf_Mold 0052	F	1	–	0.50	0.00	0.31	0.17	18.4	1.8	19.3
Tomato Leaf_Mold 0052	K	1	–	0.50	0.00	0.00	0.00	10.4	1.0	14.5

Table 3: Specimen-level paired comparisons (seeds averaged within specimen). Cluster bootstrap over disease classes; * fewer than 3 classes, so the cluster CI is degenerate or unreliable (specimen-bootstrap CI shown as sensitivity). With n ≤ 9 the Wilcoxon test cannot reach p < 0.05 for n < 6; CIs and sign counts are the primary summary.

Comparison	Metric	n (classes)	mean a	mean b	diff [cluster CI]	specimen CI	a>b / a<b	Wilcoxon p	Holm p
J vs G	geometry_recall	6 (2)	0.44	0.07	+0.37 [+0.11, +0.63]*	[+0.13, +0.61]	4/0	0.125	0.875
J vs G	unsupported_rate	9 (3)	0.06	0.17	-0.10 [-0.20, +0.00]	[-0.28, +0.03]	1/3	0.375	1.000
J vs E	coherence	9 (3)	0.81	0.31	+0.50 [+0.25, +0.66]	[+0.37, +0.61]	9/0	0.004	0.043
J vs E	grain_ratio	9 (3)	1.28	12.94	-11.66 [-14.68, -6.20]	[-15.89, -7.94]	0/9	0.004	0.043
K vs E	appearance_recall	7 (3)	0.50	0.29	+0.21 [-0.50, +0.67]	[-0.14, +0.57]	2/1	0.500	1.000
K vs E	grain_ratio	15 (5)	8.71	16.98	-8.27 [-12.81, -4.00]	[-14.15, -4.24]	0/15	0.000	0.001
K vs E	fuzz_ratio	15 (5)	1.00	1.90	-0.90 [-1.00, -0.79]	[-1.01, -0.78]	0/15	0.000	0.001
H vs F	geometry_recall	6 (2)	0.56	0.00	+0.56 [+0.33, +0.78]*	[+0.22, +0.89]	4/0	0.125	0.875
H vs F	appearance_recall	3 (1)	0.67	0.67	+0.00 [+0.00, +0.00]*	[+0.00, +0.00]	0/0	–	–
H vs F	grain_ratio	9 (3)	7.09	15.73	-8.64 [-11.86, -2.93]	[-11.96, -5.45]	0/9	0.004	0.043
F vs E	appearance_recall	7 (3)	0.50	0.29	+0.21 [-0.50, +0.67]	[-0.14, +0.57]	2/1	0.500	1.000
H vs G	appearance_recall	3 (1)	0.67	0.00	+0.67 [+0.67, +0.67]*	[+0.00, +1.00]	2/0	0.500	1.000
H vs G	geometry_recall	6 (2)	0.56	0.07	+0.48 [+0.33, +0.63]*	[+0.17, +0.80]	4/0	0.125	0.875
F vs E	coherence	15 (5)	0.31	0.31	+0.00 [+0.00, +0.00]	[+0.00, +0.00]	11/4	0.030	0.241
K vs E	appearance_recall_v1	3 (1)	0.00	0.00	+0.00 [+0.00, +0.00]*	[+0.00, +0.00]	0/0	–	–
F vs E	appearance_recall_v1	3 (1)	0.00	0.00	+0.00 [+0.00, +0.00]*	[+0.00, +0.00]	0/0	–	–

Table 4: Geometry × KB-conditioning ablation (mean over specimens with all four arms).

Cell	total KB recall
D (no geo, no KB)	0.00
K (no geo, KB)	0.17
G (geo, no KB)	0.04
H (geo, KB)	0.53
interaction (H−G)−(K−D)	0.32
specimens with all four cells	6

3.4 2D baselines

Table 5: 2D arms vs the real photo (specimen-averaged). Outside-mask change = changed area outside the organ mask ÷ organ area (shape-change capacity).

Arm	n	ΔE in organ	outside change	KB colour recall	unsupported colours
B	15	32.8	0.00	0.38	0.48
C	15	47.1	0.00	0.71	0.77
C_nogeo	9	42.5	0.00	0.25	0.91
C_mismatch	15	37.4	0.00	0.29	0.85
C_stage	15	36.8	0.00	0.69	0.58
I	15	43.3	0.83	0.74	0.74

3.5 Cross-view consistency

Table 6: Correspondence-based cross-view consistency: median CIELAB ΔE of the same surface point between adjacent ring views (lower = more consistent). Internal consistency only.

Specimen	D floor	Bmv	Cmv	E	F	K	J	G	H
Citrus Canker 0004	4.0	11.9	18.6	9.6	10.2	10.5	–	–	–
Citrus Canker 0012	1.3	5.5	7.9	9.4	10.9	12.0	–	–	–
Citrus Canker 0016	2.0	12.7	15.3	9.0	9.1	10.4	–	–	–
Corn Smut 0012	1.7	11.2	11.6	8.6	9.8	8.3	1.8	1.8	8.6
Corn Smut 0025	2.3	14.0	8.0	10.0	10.8	8.1	2.3	2.4	8.7
Corn Smut 0052	3.5	17.7	8.7	9.3	10.4	9.1	3.7	3.8	9.7
Mango Anthracnose 0044	0.7	8.0	36.2	5.1	7.3	7.5	0.8	0.8	7.6
Mango Anthracnose 0049	1.0	19.6	29.5	8.5	11.2	9.9	1.0	1.1	9.8
Mango Anthracnose 0094	1.6	15.6	36.0	10.4	10.8	8.7	1.6	1.7	9.1
Peach Leaf_Curl 0006	2.4	28.6	27.6	10.6	11.8	10.0	2.9	2.3	11.2
Peach Leaf_Curl 0063	3.6	21.2	29.9	7.9	9.9	9.7	3.4	3.7	11.2
Peach Leaf_Curl 0087	2.1	21.7	32.9	7.4	9.6	9.0	2.0	2.0	9.5
Tomato Leaf_Mold 0015	2.0	16.8	18.3	10.1	10.8	8.6	–	–	–
Tomato Leaf_Mold 0052	2.2	23.8	19.3	10.6	11.0	9.2	–	–	–
Tomato Leaf_Mold 0092	1.7	19.5	24.0	8.7	9.7	7.8	–	–	–

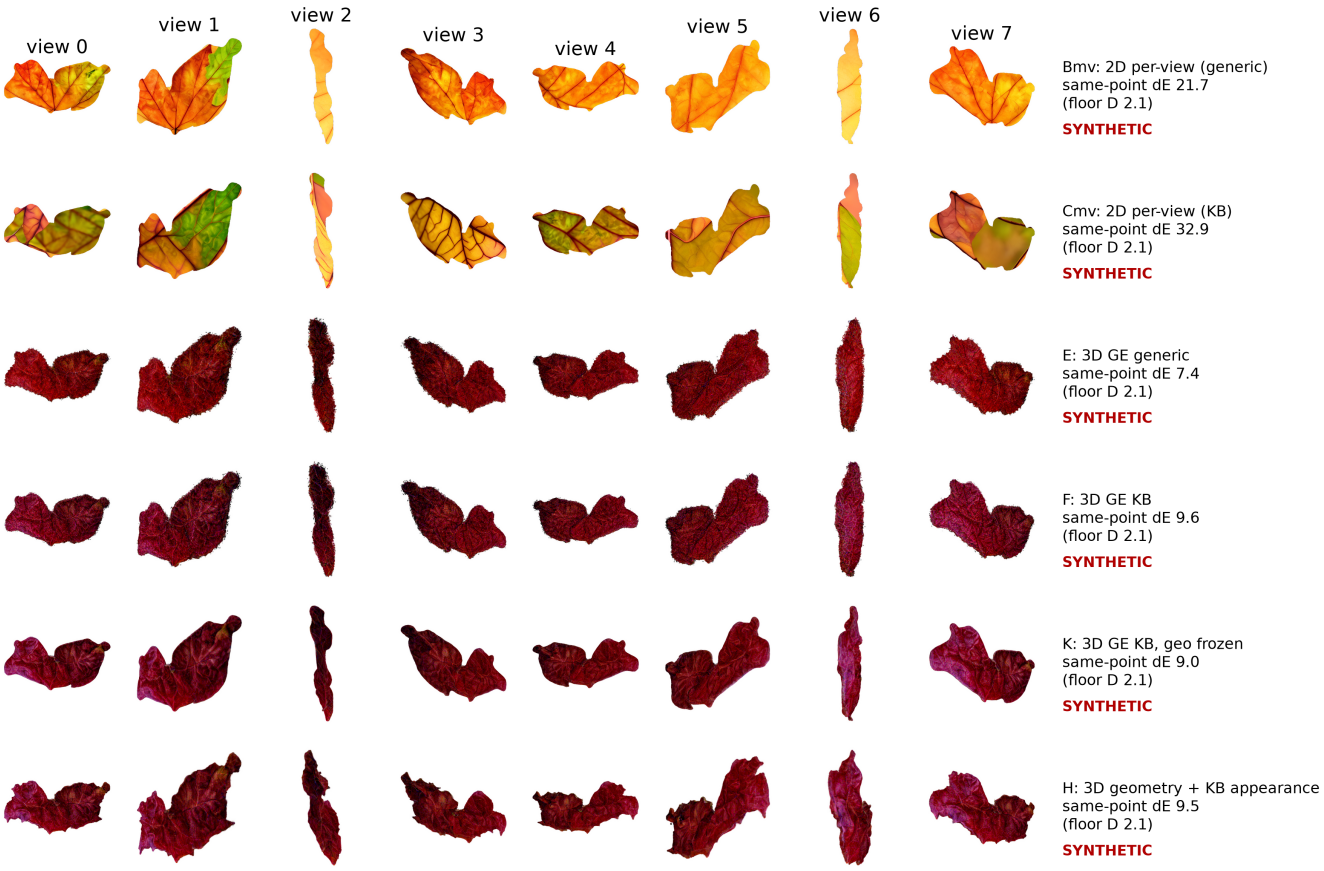

Figure 7: Per-view 2D edits vs one 3D asset (Peach 0087).

3.6 Quality control and failures

Table 7: Automated technical QC: number of flagged assets per arm (seed-level). Technical QC is not biological validation.

Flag	E	F	G	H	J	K
exploding_deformation	0/15	0/15	3/27	1/15	3/27	3/15
fuzzy_edges	13/15	12/15	0/27	2/15	6/27	0/15
ge_unverified	3/15	0/15	0/27	0/15	0/27	0/15
grainy_texture	15/15	15/15	0/27	9/15	6/27	15/15
missing_output	0/15	0/15	0/27	6/15	0/27	0/15
specimen_reconstruction_failure	1/15	1/15	3/27	1/15	3/27	1/15
unsupported_appearance	7/15	2/15	0/27	3/15	0/27	4/15
unsupported_geometry	0/15	0/15	5/27	2/15	5/27	3/15

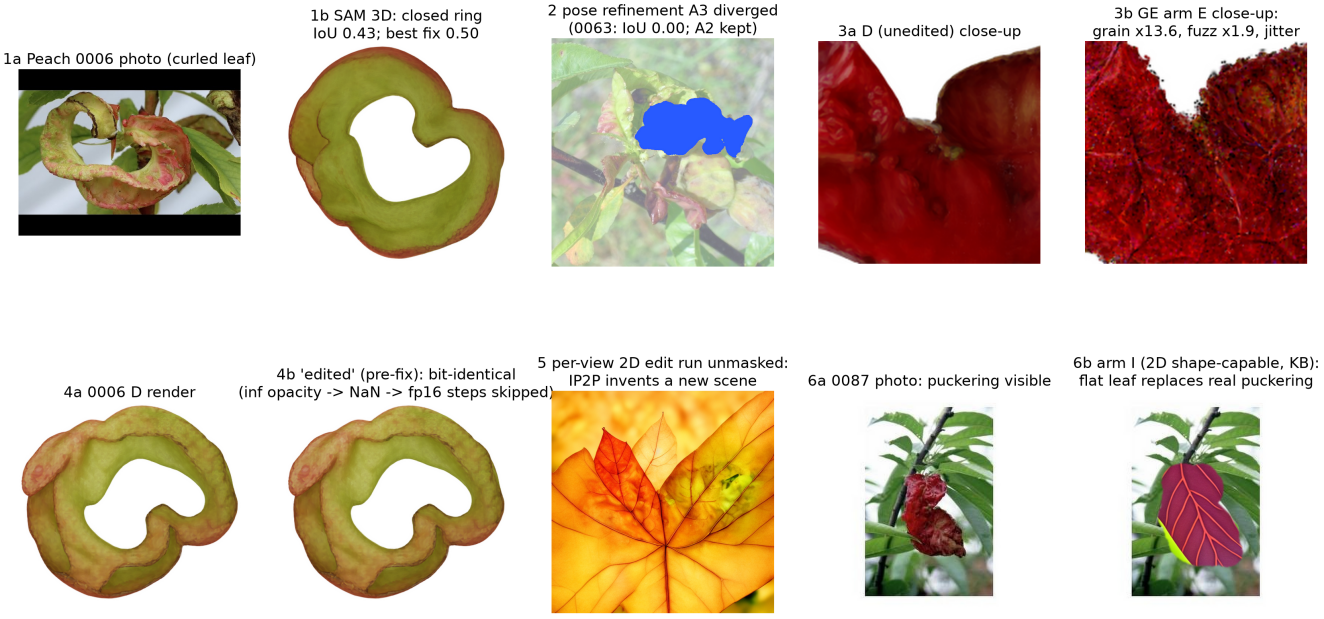

Figure 8: Failure taxonomy (real runs).

3.7 What 3D adds, and what it does not

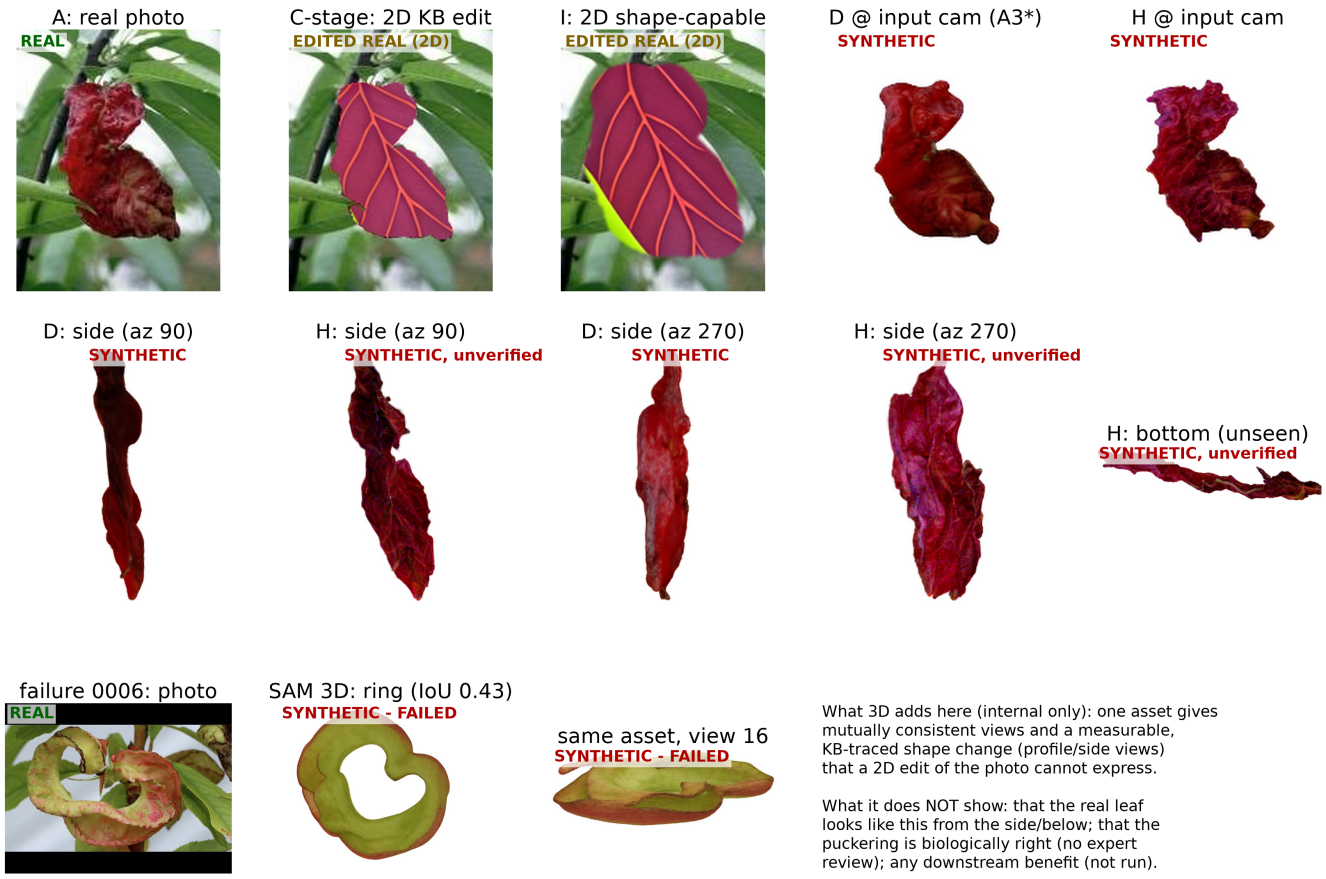

Figure 9: Added value of the 3D asset vs 2D editing, with a reconstruction failure.

4 Decision (brief §24) and next steps

Outcome A (verified 3D reconstruction): not supported. No independent reference (second view or 3D measurement) exists for any specimen. Input-view agreement after pose correction is a fit, not an accuracy measure.

Outcome B (knowledge-controlled 3D simulation): technical preconditions partly met; not established. Three preconditions hold on the pilot:

- the KB-operator geometry editor gives coherent, regularised, KB-traceable shape change;
- it realises more of the stage's KB geometric attributes than an equal-magnitude generic warp (small n, 2 classes, partly by construction);
- one 3D asset is far more cross-view consistent than independent per-view 2D edits (internal consistency only).

Outcome B additionally requires *biological validation*, which was not possible (no expert raters). The detectors are coarse; the depression detector is non-specific; on volumetric organs the outgrowth response is attenuated.

Outcome C (downstream augmentation): not tested. The downstream classifier experiment (§18) was not run. The Stage-5 gate requires a reliable controlled pilot first, and 15 assets cannot power a held-out comparison. At the measured ≈3.4 GPU-h per specimen for the full arm set, 250 assets need about 850 GPU-h locally. With only D, J and H per asset the figure is about 1 GPU-h per asset (≈250 GPU-h, ≈5 days on two TITAN X).

Classification: Outcome D (no established advantage) for now. Neither GaussianEditor nor the new pipeline has been shown to produce biologically valid disease geometry, and no advantage on real data has been measured. The evidence supports a narrower technical contribution: diagnosis and repair of the pipeline, plus a controllable, auditable KB→geometry editing method with honest metrics.

Recommended next experiments, in priority order.

1. **Blinded expert review** (2–3 plant pathologists). Rate stage-matched renders of D/J/H/K/F and 2D C-stage/I per specimen for symptom presence, stage correctness and implausible artefacts. This is the only way to move toward Outcome B.

2. **Operator coverage:** a directional *roll* operator. Grape Leafroll's text specifies "margins rolled under", the clearest geometric KB claim in SAGE. Also calibrate outgrowth on synthetic volumes, and lesion-anchored (observed-spot) placement instead of random sites.

3. **Two-sided leaves:** SAM 3D leaves are single SH-0 sheets, so lower-surface symptoms (Tomato Leaf Mold, Squash Downy Mildew) cannot be represented. Duplicate the sheet with offset normals before any underside edit.

4. **Reference data for external validity:** even 20–30 specimens photographed from several views would turn every "unverified" label into a measurement. (Out of scope by current decision.)

5. **Downstream test (C)** only after 1–3: matched synthetic budget, leakage-safe SAGE test split, real-only vs +C-stage 2D vs +J/H 3D renders.